SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL - 1: Experiments

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

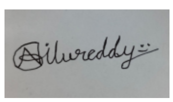
Name	**G. Venkata Niranjan Reddy**
Register Number	**192525199**

COURSE OUTCOME COVEREDCO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)
Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3) **Total Marks: 40**
Question Count: 4

Duration : 60 Minutes

Code of Conduct
I G. Venkata Niranjan Reddy (192525199) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Experiment 1:

Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

AIM: To Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS

PROCEDURE:

step1: Go to zoho.com.

step 2: Log into the zoho.com.

step 3: Select one application step.

step4: Enter application name as library book reservation system.

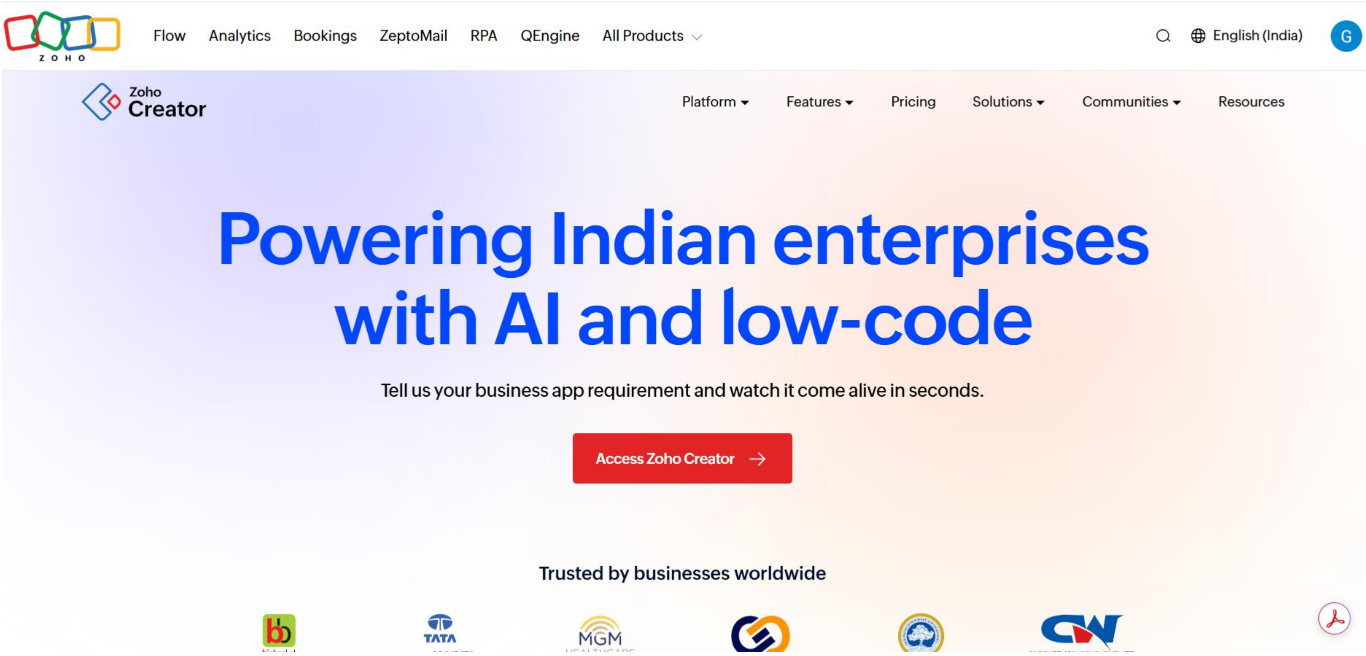
step 5: Created new application as library book reservation system.

step 6: Select one form s

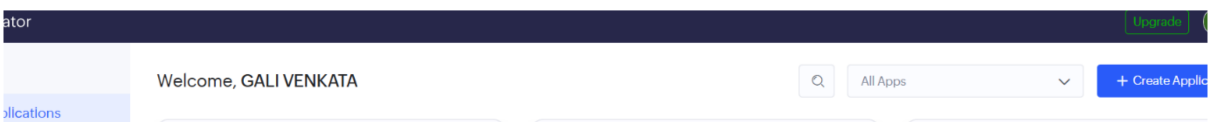
step 7: The software has been created.

IMPLEMENTATION:

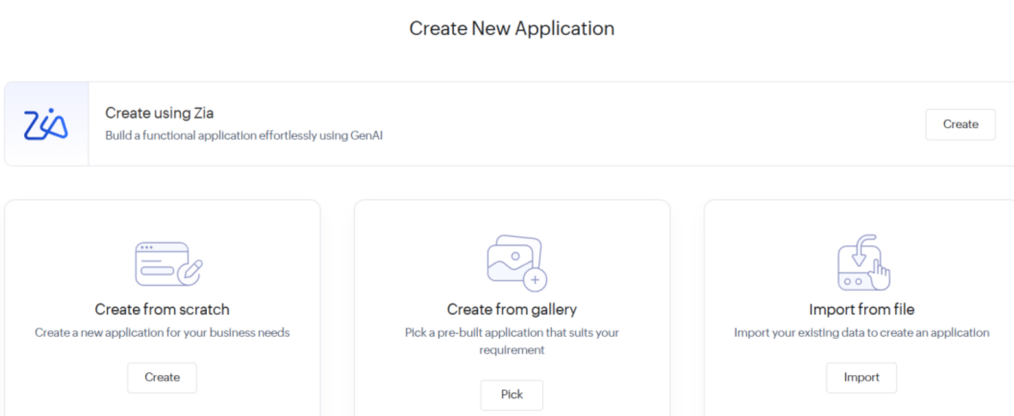
STEP1: GOTO ZOHO.COM



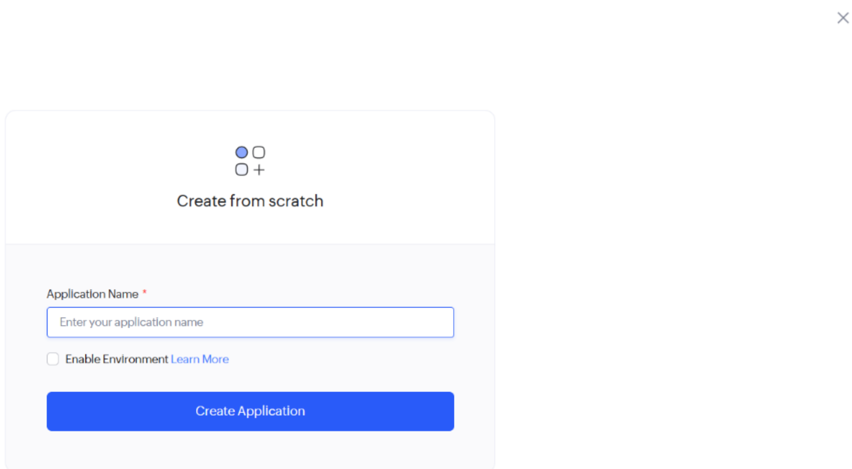
STEP 2: LOGINTOTHE ZOHO.COM



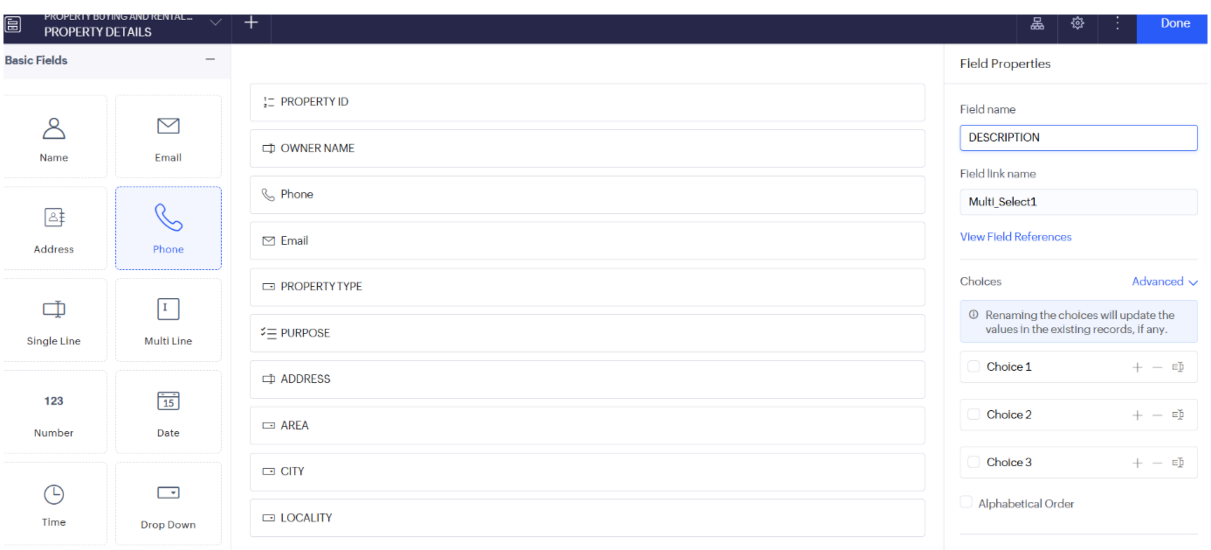
STEP 3: SELECT ONE APPLICATION



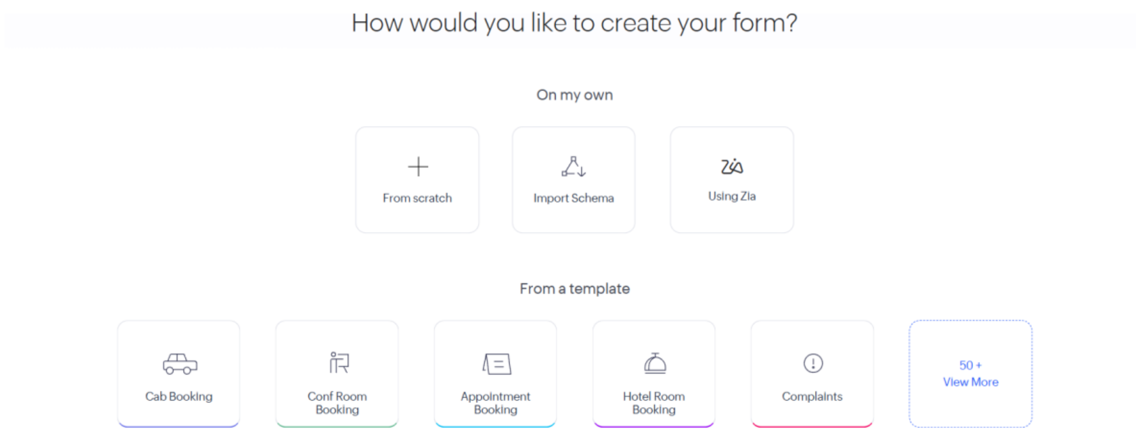
STEP 4: ENTER APPLICATION NAME



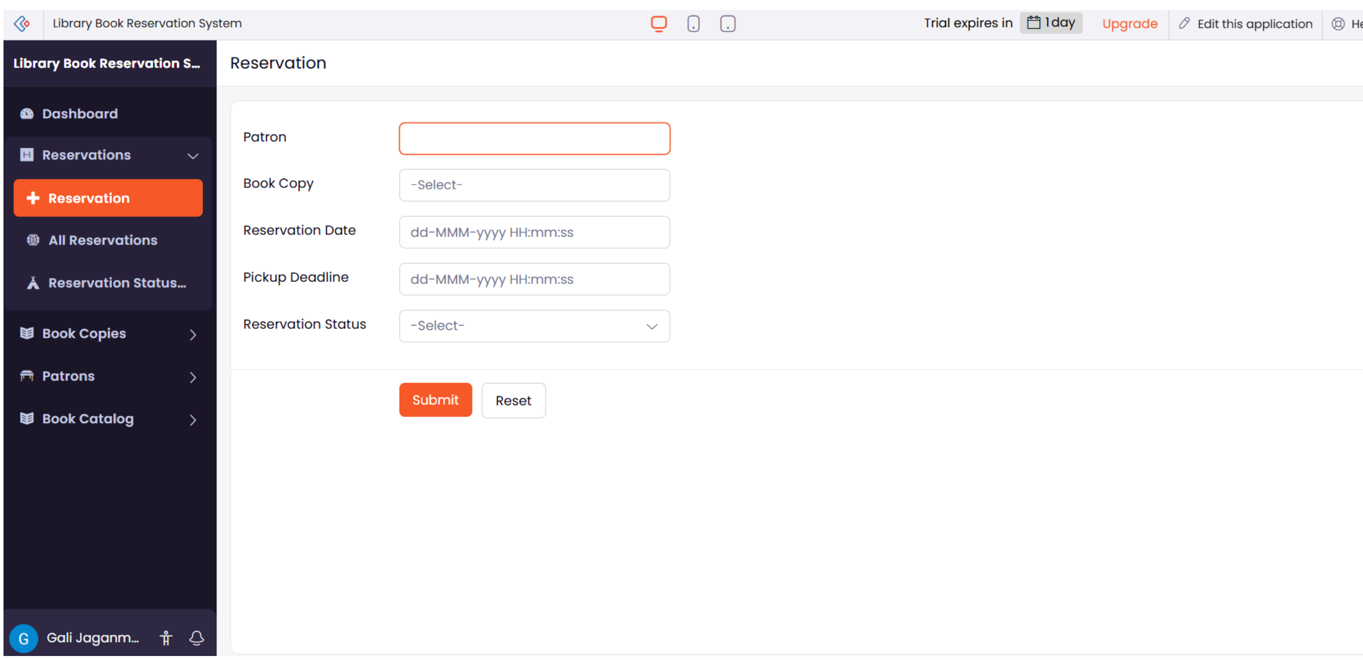
STEP 5: CREATED NEW APPLICATION

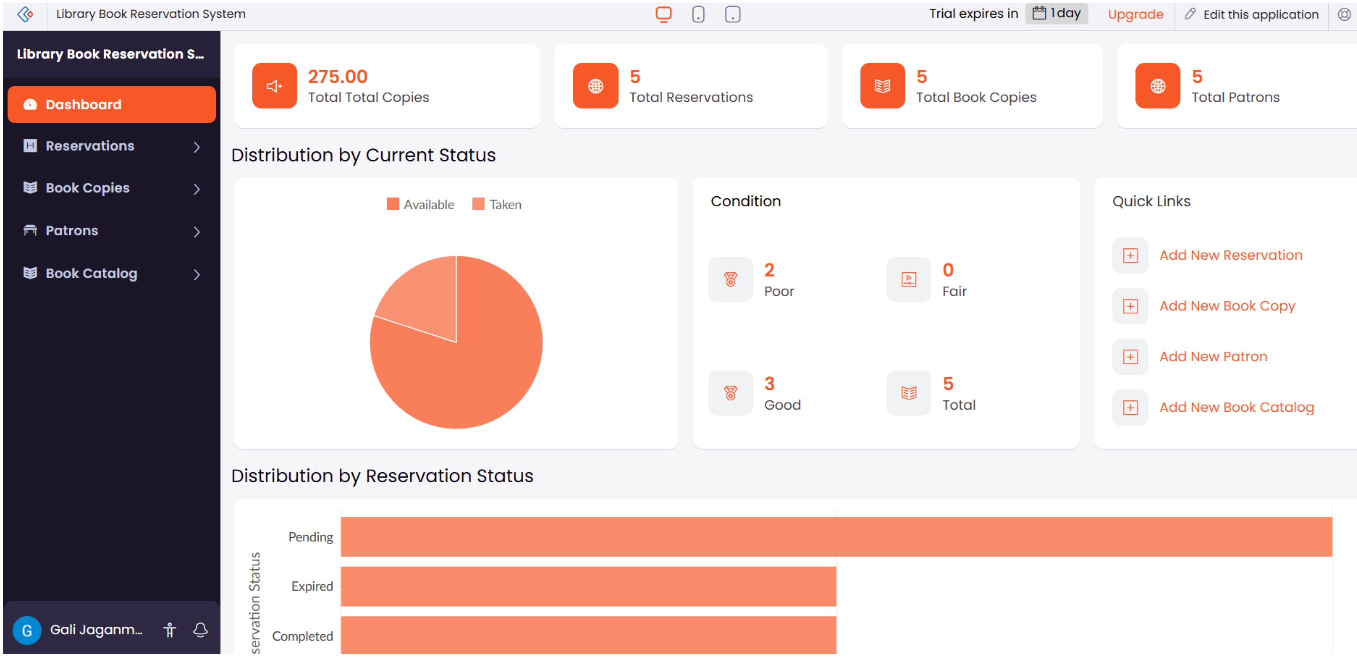


STEP 6: SELECT ONE FORM



STEP 7: THE SOFTWARE HASE BEEN CREATED.





Experiment 2:

Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

AIM:

To create a simple cloud software application for **Payroll Processing System** using **Zoho Creator** as a Cloud Service Provider to demonstrate the **Software as a Service (SaaS)** model.

PROCEDURE:

Step 1: Go to zoho.com.

Step 2: Log in to your Zoho account.

Step 3: Open Zoho Creator from the Zoho applications.

Step 4: Click Create New Application.

Step 5: Enter the application name as Payroll Processing System.

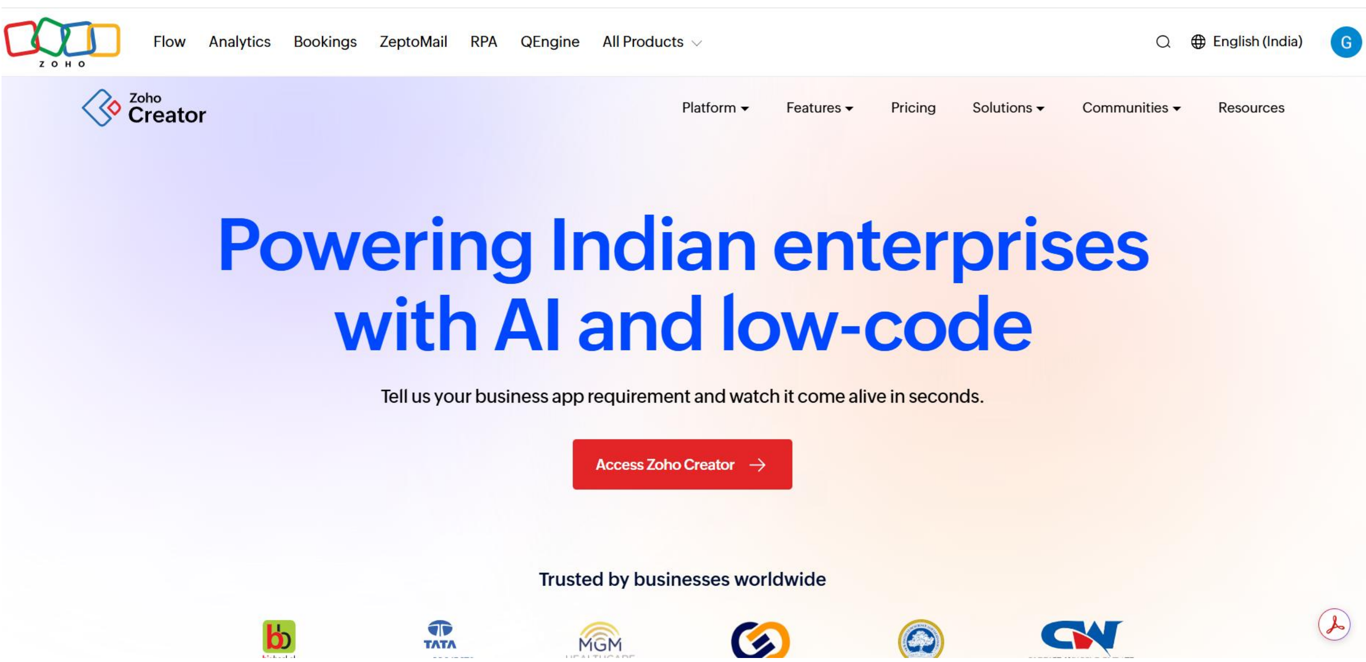
Step 6: Create a new application.

Step 7: Create a new form named Employee Payroll Details.

Step 8: The Payroll Processing System software has been successfully created and deployed on the Zoho cloud platform.

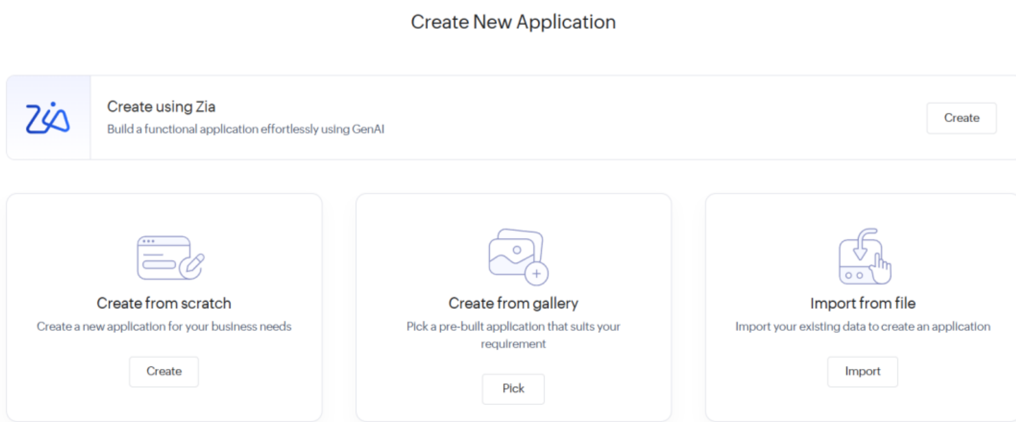
IMPLEMENTATION:

STEP1: GOTO ZOHO.COM

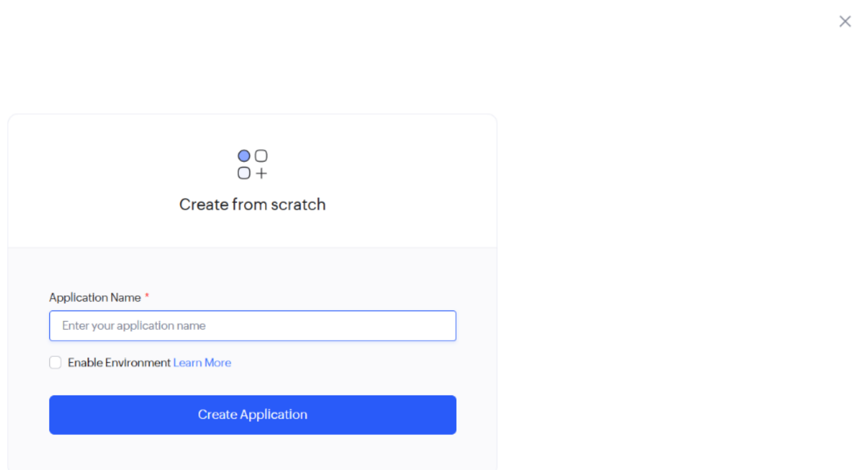


STEP 2: LOGINTOTHE ZOHO.COM

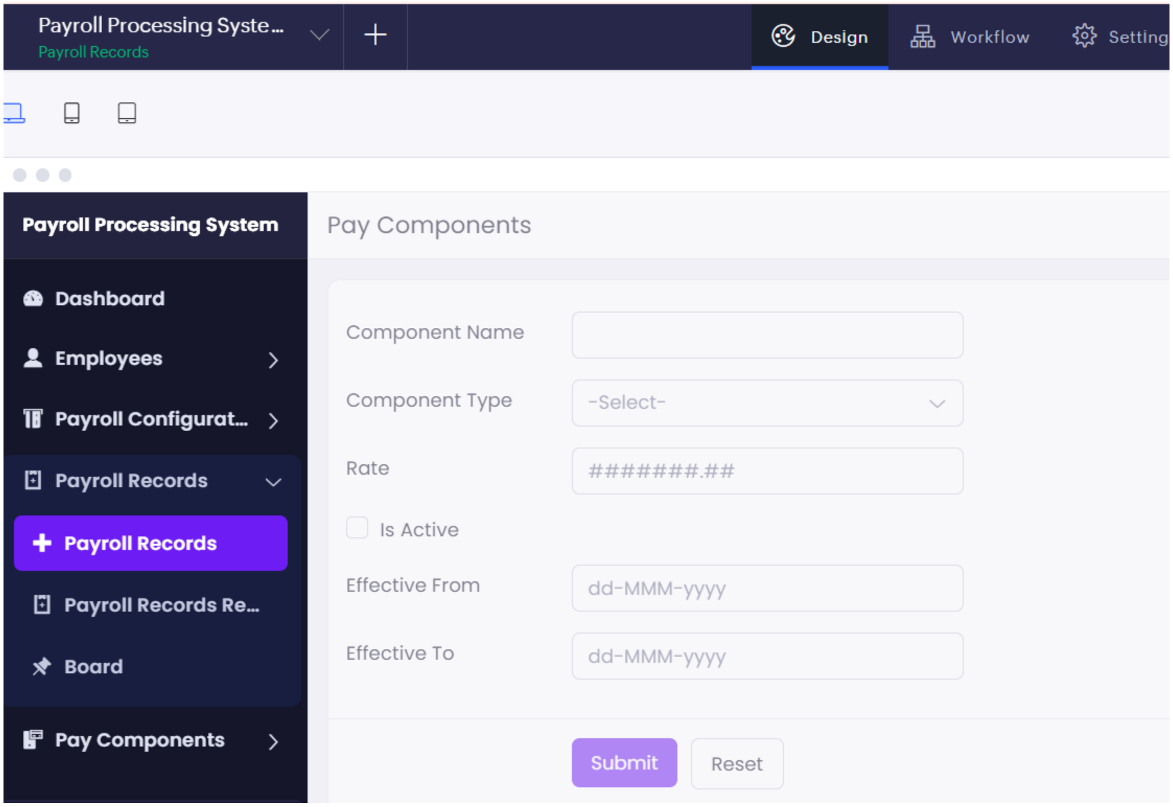
STEP 3: SELECT ONE APPLICATION



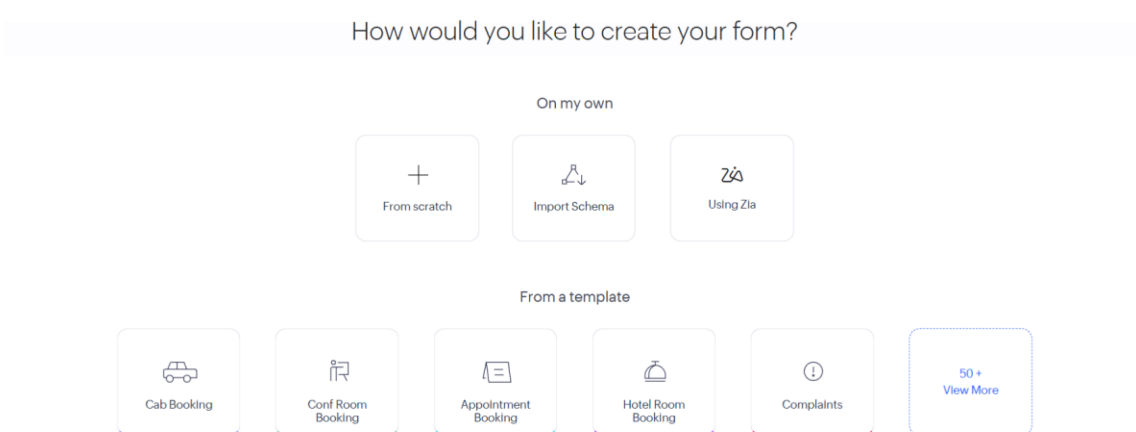
STEP 4: ENTER APPLICATION NAME



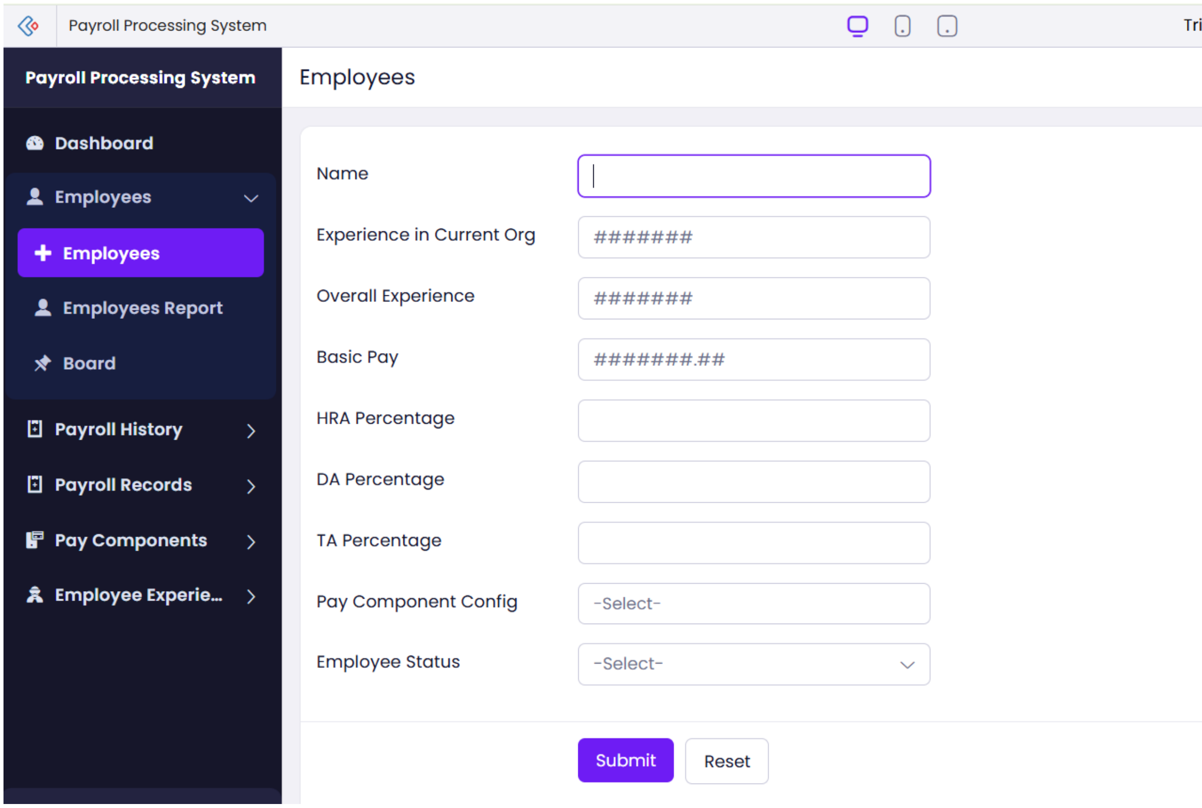
STEP 5: CREATED NEW APPLICATION

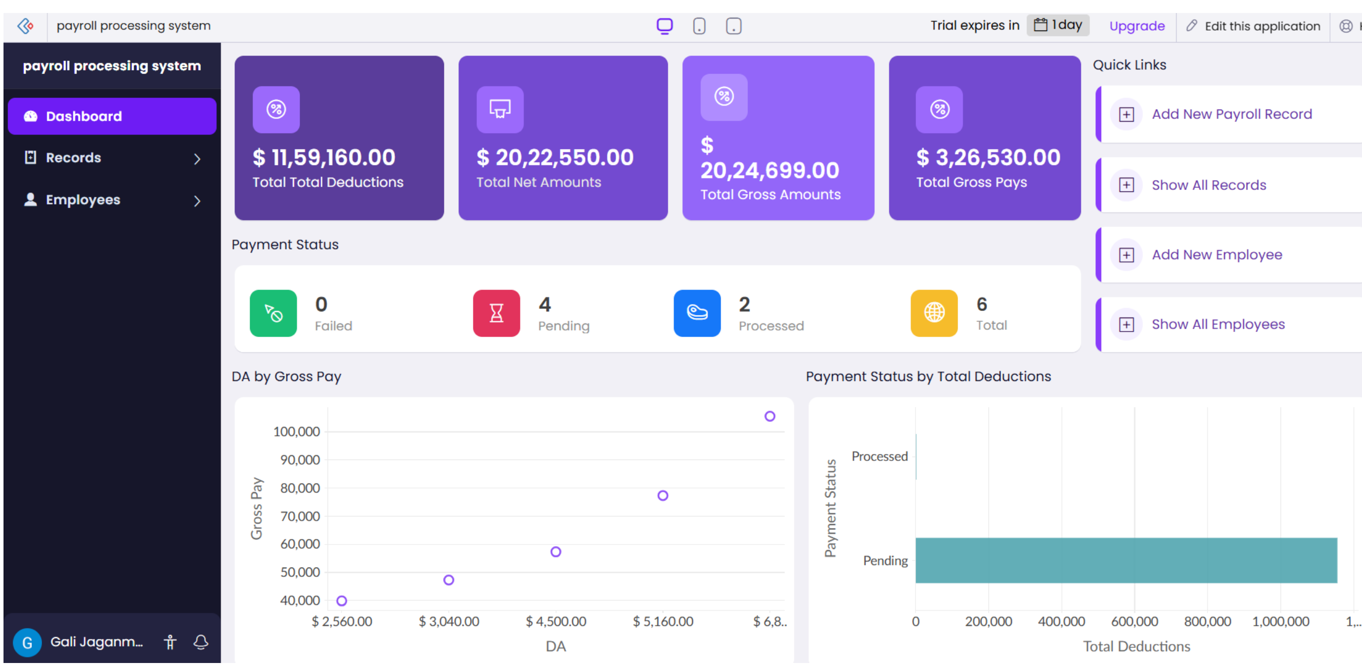


STEP 6: SELECT ONE FORM



STEP 7: THE SOFTWARE HASE BEEN CREATED.





Experiment 3:

Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Aim

To demonstrate Type-2 virtualization by installing a Type-2 hypervisor such as Oracle VirtualBox on a host operating system and creating a virtual machine with 1 CPU, 2 GB RAM, and 15 GB storage.

PROCEDURE:

STEP 1: Download VMware workstation and installed as type 2hypervisor.

STEP2: Download ubuntu or tiny OS as iso image file.

STEP 3: In VMware workstation->create new VM.

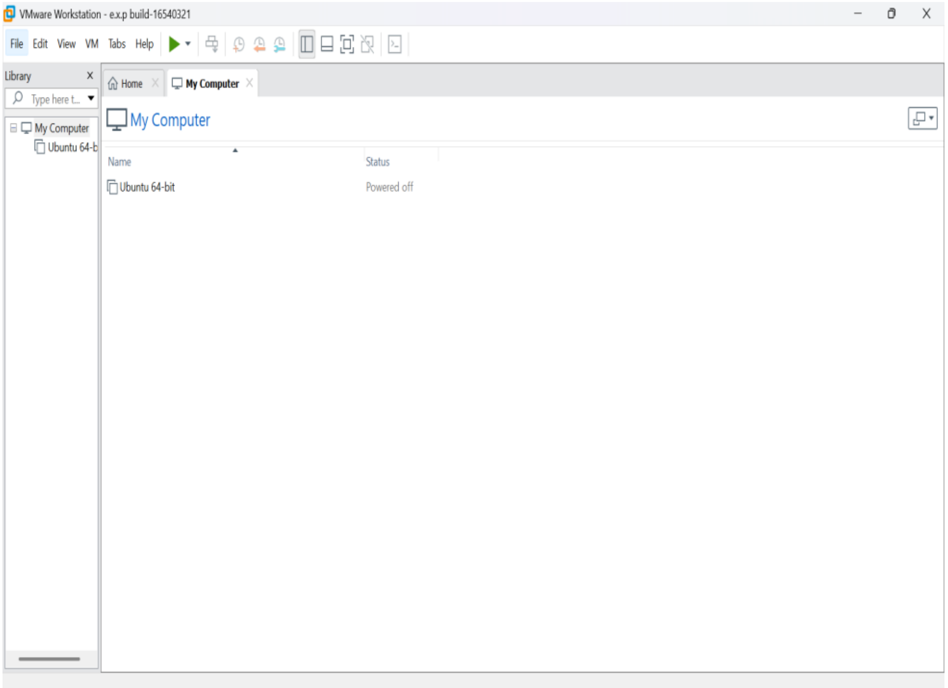
STEP 4: Do the basic configuration settings.

STEP 5: Created tiny OS virtual machine.

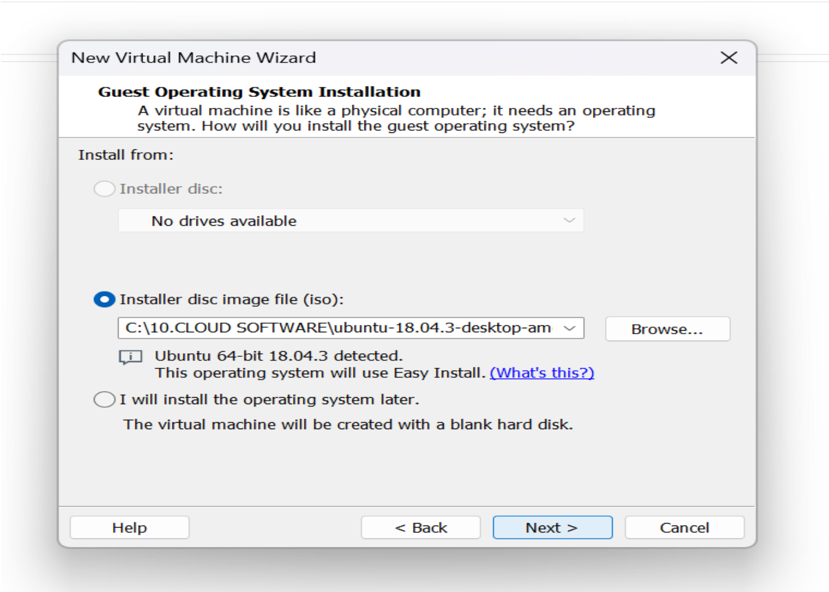
STEP 6: Launch the VM.

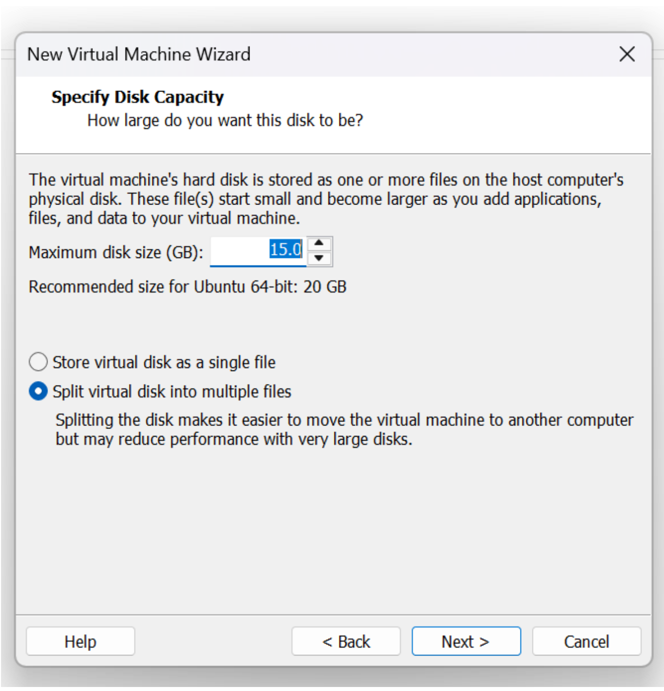
IMPLEMENTATION:

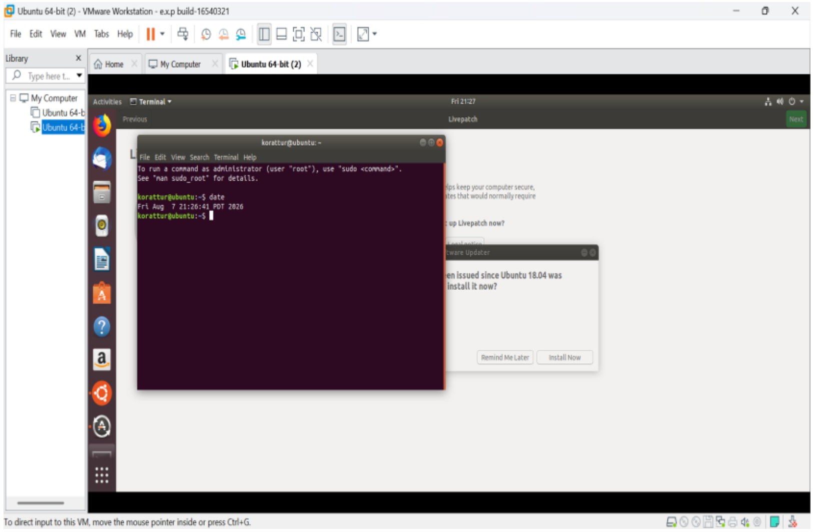
STEP 1: DOWLOAD VMWARE WORKSTATIONAND INSTALLED AS TYPE 2HYPERVISOR


VMware Workstation - e.x.p build-16540321
File Edit View VM Tabs Help
Library
Home
My Computer
My Computer
Ubuntu 64-bit
Name
Status
Powered off


New Virtual Machine Wizard
VMWARE
WORKSTATION
TECH PREVIEW™
20H2
JULY
Welcome to the New Virtual Machine Wizard
What type of configuration do you want?
Typical (recommended)
Create a Workstation Beta virtual machine in a few easy steps.
Custom (advanced)
Create a virtual machine with advanced options, such as a SCSI controller type, virtual disk type and compatibility with older VMware products.
Help
< Back
Next >
Cancel


New Virtual Machine Wizard
Guest Operating System Installation
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?
Install from:
Installer disc:
No drives available
Installer disc image file (iso):
C:\10.CLOUD SOFTWARE\ubuntu-18.04.3-desktop-am
Browse...
Ubuntu 64-bit 18.04.3 detected.
This operating system will use Easy Install. (What's this?)
I will install the operating system later.
The virtual machine will be created with a blank hard disk.
Help
< Back
Next >
Cancel


New Virtual Machine Wizard
Specify Disk Capacity
How large do you want this disk to be?
The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.
Maximum disk size (GB):
15.0
Recommended size for Ubuntu 64-bit: 20 GB
Store virtual disk as a single file
Split virtual disk into multiple files
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.
Help
< Back
Next >
Cancel



Result:

The Ubuntu virtual machine was successfully created using VMware Workstation.

Experiment 4:

Aim

To demonstrate virtualization using VMware Workstation (Type-2 Hypervisor) by creating a virtual machine, cloning it, and testing the cloned VM.

PROCEDURE:

STEP 1:Dowload VMware workstation and installed as type 2hypervisor.

STEP2:Dowload ubuntu or tiny OS as iso image file.

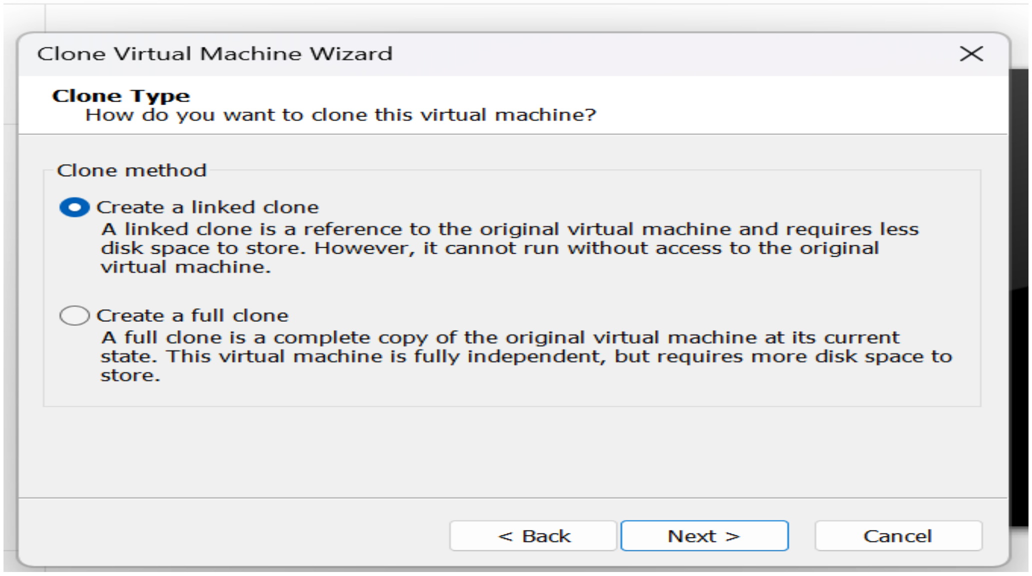
STEP 3: In VMware workstation->create new VM.

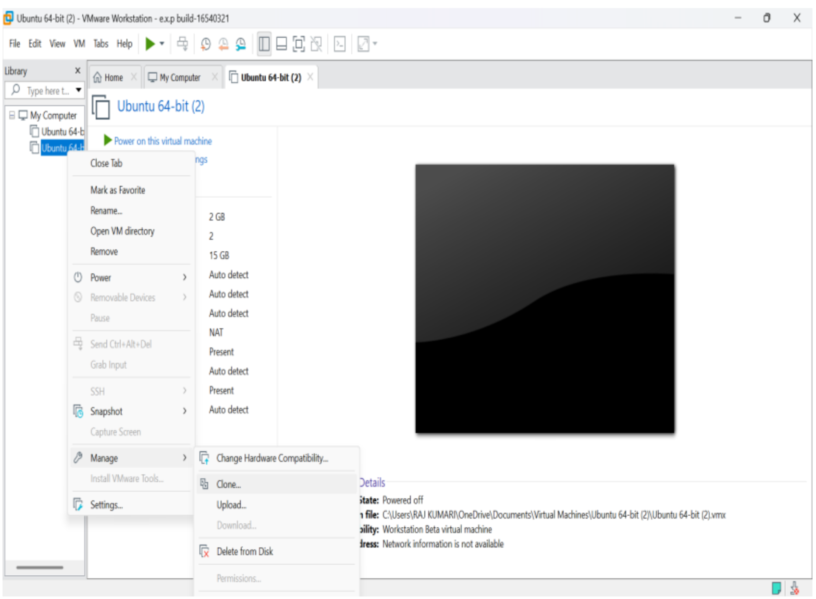
STEP 4: Do the basic configuration settings.

STEP 5: Created tiny OS virtual machine.

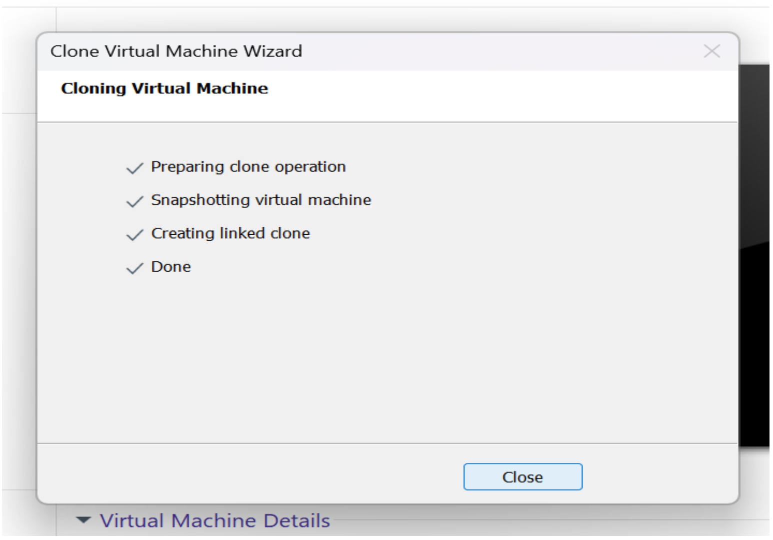
STEP 6: Launch the VM.

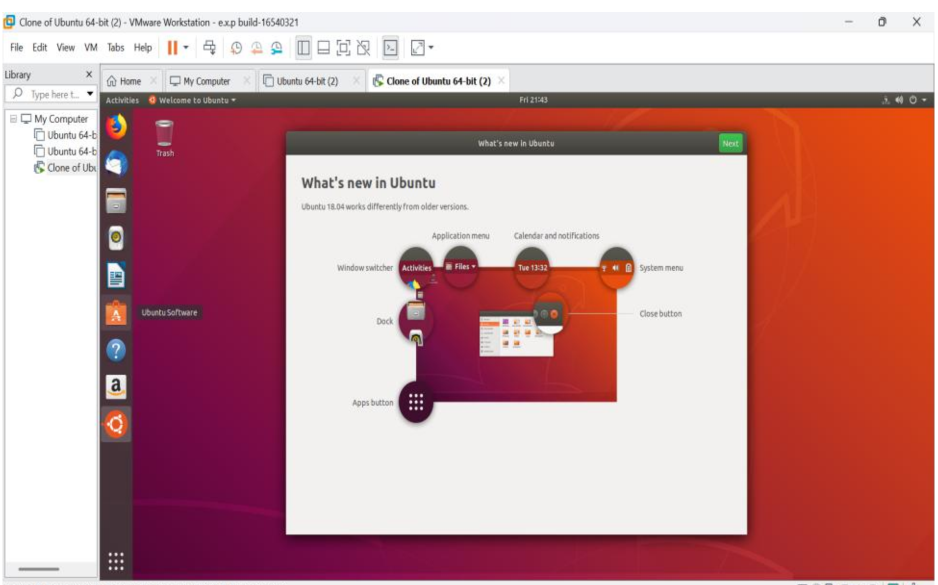
IMPLEMANTATION:






Clone Virtual Machine Wizard
VMWARE
WORKSTATION
TECH PREVIEW™
20H2
JULY
Welcome to the Clone Virtual Machine Wizard
This wizard will help you create a copy of this virtual machine.
If you intend for other users to create clones from this virtual machine, you should enable template mode under VM > Settings > Options > Advanced.
< Back
Next >
Cancel
Virtual Machine Details


Clone Virtual Machine Wizard
Cloning Virtual Machine
Preparing clone operation
Snapshotting virtual machine
Creating linked clone
Done
Close
Virtual Machine Details


Clone of Ubuntu 64-bit (2) - VMware Workstation - e.x.p build-16540321
File Edit View VM Tabs Help
Library
Home
My Computer
Ubuntu 64-bit (2)
Clone of Ubuntu 64-bit (2)
Trash
Ubuntu Software
What's new in Ubuntu
Ubuntu 18.04 works differently from older versions.
Application menu
Calendar and notifications
Window switcher
System menu
Dock
Close button
Apps button
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Result:

The Ubuntu virtual machine was successfully created using VMware Workstation.